FREEDOM INTERNATIONAL SCHOOL

WORKSHEET-SOLUTIONS

PHYSICS

ELECTRIC CHARGES AND FIELDS

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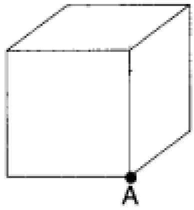
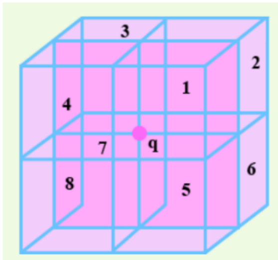
1. The total flux through the faces of the cube with side of length 'a' if a charge q is placed at corner A of the cube is



(a) $\frac{q}{8\varepsilon_0}$ (b) $\frac{q}{4\varepsilon_0}$ (c) $\frac{q}{2\varepsilon_0}$ (d) $\frac{q}{\varepsilon_0}$

Answer: Option (a)

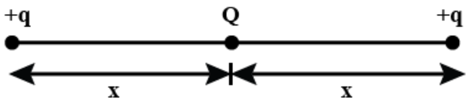
Imagine 3D picture with the charge q at the origin, and 8 cubes of side a placed such that each having one vertex at the origin, touching the charge q as shown in the figure.



∴ Flux through given cube of side a= Net flux/8

$$= \frac{q}{8\varepsilon_0}$$

2. A charge Q is placed at the centre of the line joining two point charges +q and +q is as shown in figure. The system is in equilibrium. The ratio of charges Q and q is



(a) 4 (b) ¼ (c) -4 (d) - ¼

Answer: Option (d)

For a system to be in equilibrium, net force on q=0

i.e $$\frac{Qq}{4\pi\varepsilon_0 x^2} + \frac{qq}{4\pi\varepsilon_0 (2x)^2} = 0$$

$Q = \frac{-q}{4}$ $\frac{Q}{q}$ = - ¼

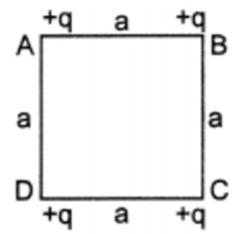
3. Two charges q_1 and q_2 are placed in vacuum at a distance d and the force acting between them is F. If a medium of dielectric constant 4 is introduced around them, the force now will be

(a) F (b) F/2 (c) F/4 (d) 4 F

Answer: Option (c)

F_{med} = F/K = F/4

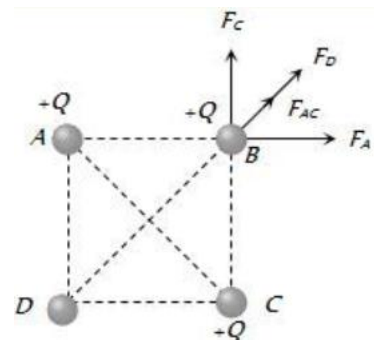
4. Four equal charges q are placed at the four corners A, B, C, D of a square of length a. The magnitude of the force on the charge at B will be



(a) $\frac{3q^2}{4\pi\varepsilon_0 a^2}$ (b) $\frac{4q^2}{4\pi\varepsilon_0 a^2}$

(c) $\frac{(1+2\sqrt{2})q^2}{2\times 4\pi\varepsilon_0 a^2}$ (d) $\frac{\left(\frac{2+1}{\sqrt{2}}\right)q^2}{4\pi\varepsilon_0 a^2}$

Answer: Option (c)



$$F_{net} = F_{AC} + F_D = \sqrt{F_A^2 + F_C^2} + F_D$$

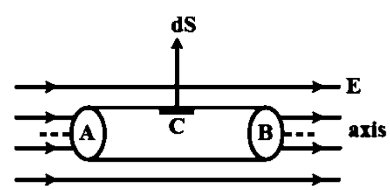
$$\text{Since} F_A = F_C = \frac{kq^2}{a^2} \text{and} F_D = \frac{kq^2}{(a\sqrt{2})^2}$$

$$F_{net} = \frac{\sqrt{2}kq^2}{a^2} + \frac{kq^2}{2a^2} = \frac{kq^2}{a^2}(\sqrt{2} + \frac{1}{2}) = \frac{q^2}{4\pi\varepsilon_0 a^2}(\frac{1+2\sqrt{2}}{2})$$

5. A cylinder of radius R and length L is placed in a uniform electric field E parallel to the cylinder axis. The total flux for the surface of the cylinder is given by

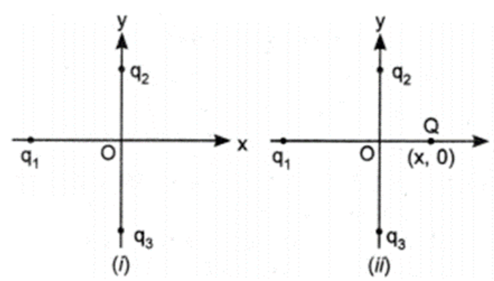
(a) $2\pi R^2 E$ (b) πR^2 (c) $\frac{\Pi R^2 - \Pi R}{E}$ (d) zero

Answer: Option (d)



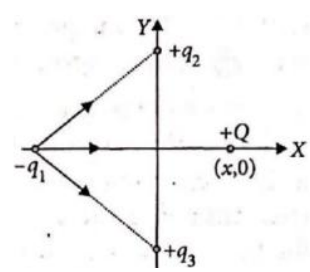
6. In Fig. (i) two positive charges q_2 and q_3 fixed along the y-axis, exert a net electric force in the +x direction on a charge q_1 fixed along the x-axis. If a positive charge Q is added at (x, 0) in figure (ii), the force on q_1

(a) shall increase along the positive x-axis.
(b) shall decrease along the positive x-axis.
(c) shall point along the negative x-axis.
(d) shall increase but the direction changes because of the intersection of Q with q_2 and q_3



Answer: Option (a)

Since positive charge q_2 and q_3 exert a net force in the +X-direction on the charge q_1 fixed along the X-axis, the charge q_1 is negative as shown in figure. Obviously, due to addition of positive charge Q at (x,0), the force on −q shall increase along the positive X-axis.



7. Flux coming out from a balloon of radius 10 cm is $1.0 \times 10^3\ Nm^2C^{-1}$. If the radius of the balloon is doubled, the flux coming out from the balloon will be ______.

(a) 0.5 times (b) 2 times (c) same (d) 4 times

Answer: Option C

8. An electric dipole coincides on the Z-axis and its midpoint is on the origin of the coordinate system. The electric field at an axial point at a distance z from the origin is E_z and electric field at an equatorial point at a distance y from the origin is E_y. Here z=y>>a, so $\frac{|Ez|}{|Ey|}$ is equal to

(a) 1 (b) 4 (c) 3 (d) 2

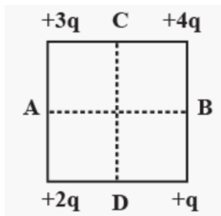
Answer: Option (d)

On the axis of dipole, $E_z=\frac{2p}{4\Pi\varepsilon_0 z^3}$

On the equatorial point of dipole, $E_y=\frac{p}{4\Pi\varepsilon_0 y^3}$

$$\frac{|Ez|}{|Ey|}=2$$

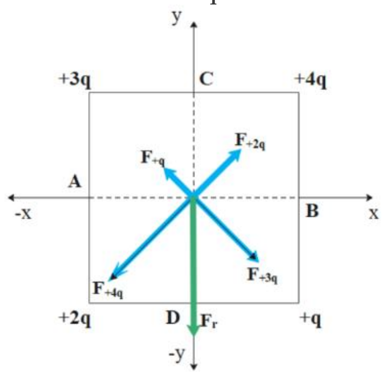
9. Four charges are arranged at the corners of a square as shown in the figure. The direction of electric field at the centre of the square is along



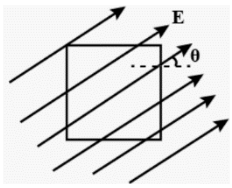
(a) CD (b) BC (c) AB (d) AD

Answer: Option (a) CD

Imagine a unit positive test charge at the centre of the square.



10. A square surface of side L meter in the plane of the paper is placed in a uniform electric field E (volt/m) acting along the same plane at an angle with the horizontal side of the square as shown in the figure. The electric flux linked to the surface, in units of volt−m, is



(a) EL^2 (b) $EL^2 \cos\Theta$

(b) $EL^2 \sin\Theta$ (d) zero

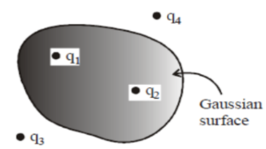
Answer: Option (d)

The angle between the area vector and the field lines is 90°.

For questions 11 to 15, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. **Both Assertion and Reason are true and Reason is correct explanation of Assertion.**
B. **Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.**
C. **Assertion is true but Reason is false.**
D. **Both Assertion and Reason are false.**

11. **Assertion:** In a non-uniform electric field, a dipole will have translatory as well as rotatory motion.
Reason: In a non-uniform electric field, a dipole experiences a force as well as torque.
Answer: Option (a)
12. **Assertion:** A metallic shield in the form of a hollow shell may be built to block an electric field.
Reason: In a hollow spherical shield, the electric field inside it is zero at every point.
Answer: Option (a)
13. **Assertion:** Four point charges q_1, q_2, q_3 and q_4 are as shown in figure. The flux over the shown Gaussian surface depends only on charges q_1 and q_2.



Reason: Electric field at all points on Gaussian surface depends only on charges q_1 and q_2.
Answer: Option (c)
14. **Assertion:** On going away from a point charge or a small electric dipole, electric field decreases at the same rate in both the cases.
Reason: Electric field is inversely proportional to square of the distance from the charge or an electric dipole.
Answer: Option (d)
15. **Assertion:** Electric force acting on a proton and an electron, moving in a uniform electric field is same, where as acceleration of the electron is 1836 times that of a proton.
Reason: An electron is lighter than a proton.
Answer: Option (a)
a=qE/m